
Classical Thermodynamics

basics

Jul 13, 2026

CONTENTS

I	Thermodynamics	3
1	Brief History of Thermodynamics	5
2	Principles of Thermodynamics	7
2.1	Lavoisier's Principle	7
2.2	First Law of Thermodynamics	8
2.3	Gibbs: Internal Energy, Phase Rule, and Multivariable Functions	8
2.4	Second Law of Thermodynamics - Clausius Statement	14
2.5	Open Systems	16
3	Thermodynamics potentials	19
3.1	First principle	19
3.2	Potentials - specific quantities	23
4	Thermodynamic coefficients	25
4.1	Thermodynamic coefficients of a single-component fluid	25
4.2	Relations between thermodynamic coefficients	25
4.3	Thermodynamic equilibrium	27
5	Stati della materia e modelli	29
5.1	Gas ideali	29
II	Thermal Engineering	31
6	Principles of Thermal Engineering	33
7	Thermodynamic transformations and thermomechanical systems	35
7.1	Turbine	35
7.2	Compressor	36
8	Thermodynamic cycles and heat engines	37
8.1	Carnot cycle and heat engine	37
8.2	Heat engines and cycles	37
8.3	Applications	39
9	Heat transfer	41
9.1	Conduction	41
9.2	Convection	42
9.3	Radiation	42

III	Electro-chemical systems	49
10	Principles of Thermodynamics - Electro-chemical systems	51
10.1	Chemical reactions	51
10.2	First principle - energy balance	51
10.3	Law of Mass Action	52
11	Nernst equation	55
12	Batteries: thermo-electro-chemical modeling	57
12.1	Daniell cell	57
12.2	Li-ion battery	61
IV	Random Notes	63
	Proof Index	65

This material is part of the [basics-books project](#). It is also available as a .pdf document.

Classical thermodynamics mainly deals with energy, its forms and the ways it can be transferred.

Introduction to thermodynamics. Principles of thermodynamics; Gibbs' formulation of classical thermodynamics: phase rule, state variables and internal energy; thermodynamic potentials and thermodynamic coefficients.

States of matter. Gas and ideal gas; mixtures and humid air; liquids; solids and elastic solids.

Thermal Engineering. Thermodynamic transformations and heat machines, converting heat and work; mechanisms of heat transfer.

Classical thermodynamics provides a macroscopic, average, description of complex systems composed by a “huge” number of elementary components, in equilibrium: equilibrium means statistical equilibrium, resulting in stationary macroscopic observable physical quantities - thermodynamic variables - even though non-stationary microscopic dynamics of the system. The development of thermodynamics is closely related with the development of chemistry and the re-born of an atomic theory of the matter. Classical thermodynamics is the results of studies on matter and energy performed from the early experiments of the XVII centuries on gas to the mathematical formulation of the late XIX century by Gibbs. A consistent theory of thermodynamics starts from conservation of mass and balance of total energy - whose changes result from input of work or heat -, provides a description of natural tendencies in nature, through Clausius' statement of the second principle - degradation of mechanical energy and heat direction from hot to cold bodies - in terms of a thermodynamic state variable, called entropy, and provides a definition of a thermodynamic scale of temperature - defined in terms of energy and entropy.

Part I

Thermodynamics

BRIEF HISTORY OF THERMODYNAMICS

See introduction to thermodynamics in the material about [physics for high-schools](#), relying on your browser translation (Italian only, so far)

- [introduction to thermodynamics](#)
- [brief history of experiments in thermodynamics](#)

PRINCIPLES OF THERMODYNAMICS

In this chapter, the principles of classical thermodynamics, along with the relevant concepts and mathematical formalism, are introduced. The principles of thermodynamics are presented for **closed systems**, and subsequently extended to open systems.

- The *principle of conservation of mass - Lavoisier's principle*, valid in classical mechanics and summarized by the formula “*nothing is created, nothing is destroyed, but everything is transformed*”, states that in a closed system, the mass is constant,

$$dM = 0 .$$

- The *first law of thermodynamics* provides the general form of the *total* energy balance of a closed system, recognizing the work done by external forces δL^{ext} and the heat δQ^{ext} exchanged between the system and the external environment as the causes of the variation of the total energy of the system.

$$dE^{tot} = \delta L^{ext} + \delta Q^{ext} .$$

- The work of **Gibbs** provides *the necessary concepts and a rigorous mathematical formalization* of classical thermodynamics. The concepts of internal energy, state variables, and Gibbs' phase rule are introduced; additionally, some phase diagrams for representing the state of a system and thermodynamic transformations are presented, which will be used in subsequent chapters.
- The *second law of thermodynamics* describes natural tendencies: the dissipation of macroscopic mechanical energy and the transfer of heat from a hot body to a cold body, in a principle expressible in terms of **entropy**,

$$dS \geq \frac{\delta Q}{T} .$$

- Finally, the balance of physical quantities for *open systems* is derived by modifying the balance equations for closed systems, introducing terms for the **flux of physical quantities due to the transport of matter** across the system's boundary.

2.1 Lavoisier's Principle

In the context of classical mechanics, the principle of mass conservation—also known as Lavoisier's principle—states that the mass M of a closed system is constant,

$$dM = 0 ,$$

that is, “*nothing is created, nothing is destroyed, everything is transformed.*”

- This principle was discovered by early chemists through the measurement of the mass of products and reactants in chemical reaction experiments.
- This principle ceases to hold in the framework of Einstein's theory of relativity, which recognizes the equivalence of mass and energy: mass and energy are two representations of a single physical quantity and are part of a balance equation. In the special case of a body at rest, this reduces to the famous expression $E = mc^2$.

2.2 First Law of Thermodynamics

The first law of thermodynamics is the total energy balance for closed systems. The variation in total energy dE^{tot} of a closed system is due to the work δL^{ext} performed on the system by external macroscopic actions and the heat δQ^{ext} transferred to the system from the outside,

$$dE^{tot} = \delta L^{ext} + \delta Q^{ext}.$$

Classical thermodynamics provides an average macroscopic description of the microscopic dynamics of a large number of elementary components (**todo** *atomic theory*). The total energy of the system can thus be interpreted as the sum of a macroscopic kinetic contribution and a microscopic content, both kinetic and potential; heat can be interpreted as the work performed on the system by microscopic actions.

2.3 Gibbs: Internal Energy, Phase Rule, and Multivariable Functions

Following Gibbs' work, this section introduces concepts such as *state variables* and *internal energy*, as well as the *Gibbs phase rule*. Later, the *first law of thermodynamics is reformulated* using the formalism introduced by Gibbs, which allows identifying the state of a system with a limited number of independent state variables and expressing the other (dependent) state variables as functions of multiple variables.

2.3.1 State Variables

Definition 2.3.1 (State Variable)

A state variable of a system is a physical property of the system that depends exclusively on the current state of the system.

Example 2.3.1 (State Variables and Non-State Variables)

State variables include temperature, pressure, internal energy, entropy, etc. Non-state variables include work or heat exchanged by the system. **todo**

2.3.2 Internal Energy

Definition 2.3.2 (Internal Energy)

The internal energy of a system is defined as the difference between the total energy and the macroscopic kinetic energy of the system,

$$E = E^{tot} - K .$$

It is possible to derive a balance for the internal energy of a closed system by subtracting the balance of kinetic energy described by the kinetic energy theorem from the balance of total energy provided by the first law of thermodynamics,

$$\begin{aligned} dE^{tot} &= \delta L^{ext} + \delta Q^{ext} \\ dK &= \delta L^{ext} + \delta L^{int} , \end{aligned}$$

The energy balance for the internal energy of a closed system then becomes

$$dE = \delta Q^{ext} - \delta L^{int} .$$

2.3.3 Gibbs Phase Rule

Definition 2.3.3 (Phase)

A phase is defined as a portion of a chemical-physical system characterized by uniform chemical-physical (macroscopic) properties.

todo

- Discussion of properties
- Examples: a mixture of miscible gases constitutes a single phase, in which the individual components cannot be macroscopically distinguished; immiscible liquids remain macroscopically separated and thus constitute multiple phases, in which distinct chemical compositions can be macroscopically identified;...

Proposition 2.3.1 (Gibbs Phase Rule)

The thermodynamic (equilibrium) state of a system is identified by a number F of independent **intensive** state variables, determined by the **Gibbs phase rule**,

$$F = C - P + 1 + W ,$$

i.e., the number of independent intensive variables (or degrees of freedom), F , of a system is a function of the number of independent components C , the number of phases P , and the number W of ways the system can manifest internal work, such as:

- internal mechanical stresses
 - contribution of surface tension
 - bond energy of the molecules of the components
 - contribution of the electromagnetic field
-

Discussion of the Gibbs Phase Rule

The equilibrium state of a system is defined by the values of the state variables, which for a non-electrically charged gas system are: temperature T , pressure p , and concentrations $C_{c,\phi}$ of the individual components $c = 1 : C$ in the individual phases $\phi = 1 : P$ within the system.

The state of the system is thus determined by the values of the $1 + W$ intensive thermodynamic variables, here $W + 1 = 2$ (T, p), and the $C P$ molar or mass fractions $n_{c,\phi}$, for a total of $N P + W + 1$ variables. In general, these variables are constrained by some conditions:

- $C(P - 1)$ phase equilibrium conditions for each component, described by the equality of chemical potentials

$$\mu_{c,\phi_1}(T, p) = \mu_{c,\phi_2}(T, p) = \dots = \mu_{c,\phi_P}(T, p)$$

- P unitarity conditions of the fractions

$$\sum_c n_{c,\phi} = 1$$

Thus, with $C P + W + 1$ variables and $P + C(P - 1) = C P - C + P$ equations, we find that the problem can be determined by

$$C P + W + 1 - C P + C - P = C - P + W + 1 = F ,$$

independent variables.

todo

- Provide examples that clarify the definition of phase (e.g., pure solids or liquids represent phases on their own), and of independent component (e.g., chemical reactions, with no excess components, create constraints that reduce the number of independent substances, thanks to stoichiometric relationships between substances)
- Discuss the role of phase fractions of a single component and the fact that they are not state variables; example phase transition from liquid to vapor: equilibrium is determined by the value of P (or T), and the vapor fraction is a result of other extensive variables of the system.

Example 2.3.2 (Closed system containing a single-component (or non-reactive), single-phase, electrically neutral (or not subjected to electromagnetic field))

In a system consisting of a compressible gas, single-component and single-phase (gas phase), electrically neutral, **todo other?**, the only form of internal work is related to compression, $\delta L^{int,rev} = PdV$, and thus $W = 1$. Therefore, the system requires

$$F = C - P + 1 + W = 1 - 1 + 1 + 1 = 2 ,$$

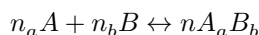
state variables to define the system state.

Example 2.3.3 (Open system containing a single-component (or non-reactive), single-phase, electrically neutral (or not subjected to electromagnetic field))

In an open system, the variation in energy of the system also depends on the variation in the amount of gas contained in it. Thus, in general, there are $W = 2$ ways to change the system's energy: by compression work or by a flow of matter into the system. Therefore, $F = 3$ state variables are needed to define the system's state.

Example 2.3.4 (Reactive gas mixture in a closed system)

In a reactive gas mixture consisting of two compounds A, B in equilibrium according to the equilibrium reaction



the energy of the system depends on the mechanical compression work of the gas mixture and the quantities of the 3 compounds present in the gas. However, the variation of these compounds is not independent but determined by the equilibrium reaction. Specifically,

$$\begin{aligned} dn_B &= dn_A \frac{n_b}{n_a} \\ dn_{A_a B_b} &= -dn_A \frac{n}{n_a} \end{aligned}$$

The reaction is thus determined by a single parameter. The variation in energy of the system is thus determined by $W = 2$ processes: from the compression work done on the system, and from the state of the reaction. To define the state of the system, $F = 3$ independent state variables are needed, such as **todo** T, P, n_A ? Are the chemical potentials $\mu_A, \mu_B, \mu_{A_a B_b}$ required? Are they uniquely determined?

Example 2.3.5 (Single-component system during a phase transition)

First-order phase transition. During a first-order phase transition, two phases, $P = 2$, are simultaneously present in the system. According to the Gibbs phase rule, the state of the system is determined by

$$F = C - P + 1 + W = 1 - 2 + 1 + 1 = 1,$$

state variable.

Critical point. The critical point in the phase diagram of a single-component system defines the condition where three phases, $P = 3$, are simultaneously present in the system. According to the Gibbs phase rule, the state of the system is determined by $F = 0$ state variables: the state of the system at the critical point is uniquely defined, with no degrees of freedom.

Example 2.3.6 (Solid)

In the absence of other physical phenomena, the only form of work in a solid is related to deformation work, $\delta L^{int,rev} = \sigma_{ij} d\varepsilon_{ij} dV$. The strain tensor is second-order and symmetric, and therefore has 6 independent components in 3-dimensional space, so $W = 6$

Example 2.3.7 (Solid mixtures)

- Phases in solid mixtures **todo**
-

Example 2.3.8 (Influence of the electromagnetic field)

- Electric field and magnetization **todo**
-

2.3.4 First Law in terms of state variables

Internal energy is an extensive variable of a thermodynamic system. In general, it can be written as a function of extensive variables that represent the ways the system manifests its internal energy (**todo** *due to work done on it, heat transferred to the system, and its chemical composition, thus the energy contained in bonds*).

$$E(S, X_k)$$

where X_k are the state variables whose variations are associated with reversible internal work, and S is the state variable whose variation is associated with heat exchange with the external environment and internal dissipative actions. **todo** *referring to the chapter on functions and multivariable calculus*

Assuming the function E is continuous and differentiable, at least piecewise, the exact differential of internal energy can be written as a function of the increments of the independent variables:

$$dE = \left. \frac{\partial E}{\partial S} \right|_{\mathbf{x}} dS + \left. \frac{\partial E}{\partial X_k} \right|_S dX_k = T dS + \sum_k F_k dX_k$$

where F_k are the generalized forces associated with changes in the generalized coordinates X_k , and T and S correspond to temperature and entropy as discussed further below. **todo**.

Note: As shown later, with this formalism, it is straightforward to express *the second and third laws of thermodynamics* as:

- $dS \geq \frac{\delta Q^{ext}}{T}$
 - $T \geq 0$
-

The expression for the differential of internal energy can be compared with the energy balance written in terms of heat transferred to the system and internal work:

$$dE = \delta Q^{ext} - \delta L^{int} = \delta Q^{ext} + \delta^+ D - \delta L^{int,rev}$$

where the internal work δL^{int} is recognized as the sum of a reversible contribution and a dissipative contribution, which is never negative: $\delta L^{int} = \delta L^{int,rev} - \delta^+ D$.

Since dE is an exact differential and $\delta L^{int,rev}$ is a reversible contribution, it follows that the sum of the two non-reversible contributions, $\delta U := \delta Q^{ext} + \delta^+ D$, is a reversible contribution. Comparing the two expressions for the differential of internal energy, we can associate reversible internal work with the sum of works formed as the product of generalized forces F_k and changes in the state variables X_k , and the term δU with the product $T dS$:

$$\begin{cases} -\delta L^{int,rev} &= \sum_k F_k dX_k \\ \delta U &= T dS \end{cases}$$

Temperature, T , and entropy, S

In the absence of external work done on the system, and in the absence of dissipation $\delta^+ D = 0$, we get:

$$dE^{tot} = dE = \delta Q^{ext}$$

$$dS = \frac{\delta Q^{ext}}{T}$$

Consider a closed and isolated system made of two subsystems in equilibrium, which can exchange heat but not work.

The total energy of the system is constant, $E = E_1 + E_2$. If the two subsystems are not initially at the same temperature, we observe a heat flow from the hotter system to the colder one, which satisfies the inequality:

$$\frac{\delta Q_{12}}{T_1} + \frac{\delta Q_{21}}{T_2} \geq 0 \quad \rightarrow \quad dS_1 + dS_2 \geq 0$$

The quantity $S = S_1 + S_2$ is non-decreasing.

2.3.5 Second and Third Laws of Thermodynamics

The formalism introduced in this section allows for a rather natural formulation of the second law and a version of the third law of thermodynamics.

This formulation of the third law of thermodynamics states that the thermodynamic temperature is always positive:

$$T := \left. \frac{\partial E}{\partial S} \right|_{\mathbf{x}} > 0$$

todo Add some words about the significance, in terms of molecular agitation and probability

In cases where this form or consequence of the third law of thermodynamics holds, the second law of thermodynamics is a consequence of the non-negativity of dissipation and the heat transfer mechanism, as will be discussed in more detail in the discussion of *composite systems*.

In the general case of a **simple system**, using the definition $dS = \frac{\delta U}{T} = \frac{\delta Q^{ext} + \delta^+ D}{T}$, the non-negativity of dissipation, $\delta^+ D \geq 0$, implies:

$$dS \geq \frac{\delta Q^{ext}}{T}$$

This is an expression of the *Clausius statement of the second law of thermodynamics*.

todo oss The third law of thermodynamics: 1. It seems not to be a fundamental principle; 2. For some systems with limited energy, the definition of temperature $T := \left(\frac{\partial E}{\partial S} \right)$ produces a negative temperature **todo** add a section on statistical mechanics?

From L.E. Reichl, *A Modern Course in Statistical Physics*, with some inconsistencies **todo** check!

$$\delta W = -PdV + JdL + \sigma dA + V \left(\vec{e} \cdot d\vec{p} + \vec{h} \cdot d\vec{m} \right) + \phi dq$$

- where J, σ are tensions per unit length and area, dL, dA are changes in length or area,
- where $\vec{e}, \vec{h}$ are the electric and magnetic fields, $\vec{p}, \vec{m}$ are polarization and magnetization,
- ϕ is the electric potential, and q is electric charge (for open systems, otherwise $dq \equiv 0$ or net charge would be created!)

Example 2.3.9 (Closed Monocomponent Gas System)

The energy of the system, $E(S, V)$

$$dE = T dS - P dV$$

Example 2.3.10 (Open Monocomponent Gas System)

The energy of the system, $E(S, V, N)$

$$dE = T dS - P dV + \mu dN$$

Example 2.3.11 (Reactive Gas Mixture in a Closed System)

The energy of the system

$$\begin{aligned} dE &= T dS - P dV + \mu_k dN_k = \\ &= T dS - P dV + (\mu_k n_k) dN \end{aligned}$$

where n_k are the stoichiometric coefficients (with sign) of the reaction, and N is a quantity that identifies the equilibrium of the reaction, such that the variation of each component can be written as $dN_k = n_k dN$.

Example 2.3.12 (Monocomponent Mixture During a Phase Transition)

todo how to treat the phase fractions?

Example 2.3.13 (Solid)

Solid system with initial volume V under uniform stress and strain, with small deformations

$$dE = T dS - V \sigma_{ij} d\varepsilon_{ij}$$

Example 2.3.14 (Solid Mixtures)

todo

Example 2.3.15 (Influence of the Magnetic Field)

$$dE = T dS - P dV + H dM$$

todo

2.4 Second Law of Thermodynamics - Clausius Statement

The Clausius statement of the second law of thermodynamics can be formulated quite naturally using the formalism introduced. There are two other famous statements of the second law of thermodynamics, the Planck and Kelvin statements, which will be presented in the context of heat engines.

2.4.1 Simple Systems

The elementary variation of entropy dS of a simple closed system at uniform temperature T is greater than or equal to the ratio of the elementary heat flux introduced into the system and the temperature of the system itself,

$$dS = \underbrace{\frac{\delta^+ D}{T}}_{\geq 0} + \frac{\delta Q^{ext}}{T} \geq \frac{\delta Q^{ext}}{T} .$$

This is the Clausius statement of the second law of thermodynamics for simple systems with uniform temperature.

2.4.2 Composite Discrete Systems

todo definition of a composite system. Heat conduction occurs between subsystems.

Entropy in classical thermodynamics is an extensive physical quantity: the entropy of a system composed of N simple subsystems is the sum of the entropies of the subsystems,

$$S = \sum_{n=1:N} S_n .$$

The entropy balance of a single subsystem that exchanges heat with the other subsystems and the external environment is written as

$$\begin{aligned} dS_i &= \frac{\delta Q_i^{ext,i}}{T_i} + \frac{\delta^+ D_i}{T_i} = \\ &= \frac{\delta Q_i^{ext}}{T_i} + \frac{\sum_{k \neq i} \delta Q_{ik}}{T_i} + \frac{\delta^+ D_i}{T_i} \geq \\ &\geq \frac{\delta Q_i^{ext}}{T_i} + \frac{\sum_{k \neq i} \delta Q_{ik}}{T_i} . \end{aligned}$$

The entropy balance of the entire system is obtained by summing the entropy balances of the individual subsystems,

$$\begin{aligned} dS &= \sum_i dS_i \geq \\ &\geq \sum_i \left\{ \frac{\delta Q_i^{ext}}{T_i} + \frac{\sum_{k \neq i} \delta Q_{ik}}{T_i} \right\} = \\ &= \sum_i \frac{\delta Q_i^{ext}}{T_i} + \underbrace{\sum_{\{i,k\}} \delta Q_{ik} \left(\frac{1}{T_i} - \frac{1}{T_k} \right)}_{\geq 0} \geq \\ &\geq \sum_i \frac{\delta Q_i^{ext}}{T_i} . \end{aligned}$$

Using the relation that represents the natural tendency of heat transfer “from a system at a higher temperature to a system at a lower temperature,”

$$\delta Q_{ik} \left(\frac{1}{T_i} - \frac{1}{T_k} \right) \geq 0 .$$

todo add reference to the natural tendency in heat transfer

2.4.3 Increase of Entropy in the Universe

If we consider the universe as the closed and isolated system (but is this true? Who knows? Maybe it's reasonable for now, but many things that seem reasonable today might be nonsense in a few years) consisting of a system of interest sys and the external environment env .

The variation of entropy in the universe is the sum of the variation in the system and in the external environment. We denote by $\delta Q_{sys,env}$ the heat flux which, if positive, increases the energy of the system and decreases that of the external environment. Assuming that the two subsystems are internally homogeneous,

$$\begin{aligned} dS^{univ} &= dS^{sys} + dS^{env} = \\ &= \frac{\delta Q_{sys,env}}{T^{sys}} + \frac{\delta Q_{env,sys}}{T^{env}} = \\ &= \frac{\delta Q_{sys,env}}{T^{sys}} - \frac{\delta Q_{sys,env}}{T^{env}} = \\ &= \delta Q_{sys,env} \left(\frac{1}{T^{sys}} - \frac{1}{T^{env}} \right) \geq 0, \end{aligned}$$

we obtain the relation

$$dS^{univ} \geq 0,$$

which predicts the “non-decrease” of the entropy of the universe.

2.4.4 Composite Continuous Systems

2.5 Open Systems

In general, the balance equation for a physical quantity in an open system is derived from the balance of the same physical quantity for a closed system, adding the contribution of the flux terms of the desired physical quantity through the system's boundary. Thus, if the balance of the physical quantity F for the closed system within volume V_t can be written as

$$\frac{d}{dt} F_{V_t} = R_{V_t}^e,$$

the balance of the same physical quantity for an open system identified by the (geometric) volume v_t can be written as

$$\frac{d}{dt} F_{v_t} = R_{v_t}^e - \Phi_{\partial v_t}(f),$$

where f is defined as the specific quantity of F per unit mass. The flux term through the boundary ∂v_t can be written as the sum of the flux contributions through portions $s_{k,t}$ of the surface $\partial v_t = \cup_k s_{k,t}$,

$$\Phi_{\partial v_t}(f) = \sum_{s_{k,t}} \dot{m}_k f_k,$$

where $\dot{m}_k = \rho_k v_{n,k}^{rel}$ is the mass flux through the surface $s_{k,t}$, and it is assumed that the quantity f_k is constant over the surface $s_{k,t}$, or that the average value over the surface has been considered.

Note: In case the quantity f is not uniform over the domain boundary and varies continuously, the flux term can be written as the summation of infinite terms through surfaces whose area tends to zero, via a surface integral, following the definition of a Riemann integral.

Note: A balance equation of a physical quantity for an open system also includes the balance equation of the same physical quantity for a closed system as a special case where the mass flux through the domain boundary is zero, $\dot{m}_k = 0$.

Here, we consider the balances (integrals, global of a system) of some fundamental physical quantities in classical mechanics: mass, momentum, angular momentum, and total energy.

2.5.1 Mass Balance

The mass balance, $F = M$, $f = 1$, for an open system is

$$\frac{d}{dt} M_{v_t} = - \sum_k \dot{m}_k$$

2.5.2 Momentum Balance

todo reference to or from classical mechanics

2.5.3 Angular Momentum Balance

todo reference to or from classical mechanics

2.5.4 Total Energy Balance

The total energy balance $F = E^{tot} = E + K$, $f = e^{tot} = e + \frac{|\vec{v}|^2}{2}$,

$$\frac{d}{dt} E_{v_t}^{tot} = P_{v_t}^{ext} + \dot{Q}_{v_t}^{ext} - \sum_k \dot{m}_k e_k^{tot}.$$

The power of external actions $P_{v_t}^{ext}$ can be written as the sum of contributions on the surfaces of the system's boundary through which there is mass flow, and impermeable surfaces that can be used to extract work from the system. If the effects of viscous stresses on the surfaces through which there is mass flow can be neglected, the power of the actions on the system can be written as

$$\begin{aligned} P^{ext} &= P^{ext, mech} + P^{ext, \Phi} = \\ &= P^{ext, mech} - \sum_k \dot{m}_k \frac{P_k}{\rho_k}, \end{aligned}$$

and the total energy balance of the system becomes

$$\frac{d}{dt} E_{v_t}^{tot} = P_{v_t}^{ext, mech} + \dot{Q}^{ext} - \sum_k \dot{m}_k h_k^{tot},$$

where the specific total enthalpy is defined as $h^{tot} = e^{tot} + \frac{P}{\rho} = e + \frac{P}{\rho} + \frac{|\vec{v}|^2}{2}$.

Example 2.5.1 (Turbine)

Example 2.5.2 (Compressor)

Example 2.5.3 (Combustion Chamber)

THERMODYNAMICS POTENTIALS

In this section, principles and the mathematical formalism of classical thermodynamics are reviewed.

3.1 First principle

For an extensive system with uniform intensive pro

$$dE = T dS + \mathbf{F} \cdot d\mathbf{X} ,$$

being E the internal energy, T temperature, S entropy, $\mathbf{F}$ generalized force, $\mathbf{X}$ generalized displacement.

Following Gibbs' formulation, internal energy E can be written as a function of a limited set of independent state variables,

$$E(S, \mathbf{X}) .$$

Internal energy E , entropy S and the generalized displacement $\mathbf{X}$ are **extensive** physical quantities, and thus (**thus?**) the derivatives

$$T := \left. \frac{\partial S}{\partial E} \right|_{\mathbf{X}} , \quad \mathbf{F} := \left. \frac{\partial E}{\partial \mathbf{X}} \right|_S$$

are **intensive** quantities. A discussion about the difference between the concept of *additivity* and *extensivity*¹. Beside being extensive, internal energy in classical thermodynamics is an homogeneous function² of order 1 of its arguments, namely

$$E(a S, a \mathbf{X}) = a E(S, \mathbf{X}) .$$

Euler's theorem for homogeneous functions holds.

Theorem 3.1.1 (Euler's theorem for homogeneous functions)

Let $f(x_i)$ and homogeneous function of order m , i.e.

$$f(a x_i) = a^m f(x_i) . \tag{3.1}$$

It follows that

$$f(x_k) = m x_i \frac{\partial f}{\partial x_i}(x_k) .$$

¹ H.Touchette, When is a quantity additive, and when is it extensive?

² <https://physics.stackexchange.com/q/677855>

Proof

Proof immediately follows, evaluating the derivative of (3.1) w.r.t. a , and evaluating for $a = 1$, i.e.

$$x_i \frac{\partial f}{\partial x_i}(a x_k) = m a^{m-1} f(x_k),$$

and for $a = 1$,

$$f(x_k) = x_i \frac{\partial f}{\partial x_i}(x_k).$$

Thus, internal energy can be written as

$$E(S, \mathbf{X}) = T S + \mathbf{F} \cdot \mathbf{X}.$$

(classical-thermodynamics:potentials:first-principle:examples)

3.1.1 First principle for different systems

- Single-component fluid, $E(S, V, N)$, i.e. internal energy E as a function of the entropy S , the volume of the system V and the number of moles N of the substance in the system

$$dE = T dS - P dV + \mu dN$$

As an alternative to the number of moles, it's possible to use the mass of the system $M = N M_m$ as an independent variable, being M_m the molar mass of the substance.

- Multi-component fluid, $E(S, V, N_k)$, i.e. internal energy E as a function of the entropy S , the volume of the system V and the number of moles N_k of all the substances in the system

$$dE = T dS - P dV + \sum_k \mu_k dN_k$$

where the change of number of particles (or moles, it depends on the description - anyway a non-dimensional number) dN_k is governed by the stoichiometric ratios of the reactions occurring in the system. As an alternative to the moles of the substances in the system, it's possible to use the total number of moles $N = \sum_k N_k$ and the molar concentrations $x_k = \frac{N_k}{N}$ as independent variables, with the algebraic constraint $\sum_k x_k = 1$, by the very definition of the molar concentrations; it's also possible to use the mass of the system and the mass concentration of the substances, $M = \sum_k M_k = \sum_k N_k M_{m,k}$, and $y_k = \frac{M_k}{M}$, with the algebraic constraint $\sum_k y_k = 1$.

- Single-component solid, $E(S, \mathbf{X})$

$$dE = T dS + \mathbf{F} \cdot d\mathbf{X},$$

in the limit of small-displacement regime, $dE = T dS + V \sigma : d\epsilon$.

Internal energy of an elastic medium

See also:

- [Small displacement:Statics:Thermodynamics](#) for thermodynamics of elastic media with thermal dilation, for unit-volume
 - There's a version for 1-dimensional systems in [Termodinamica:Stati della materia:Solidi elastici](#), **todo**
 - * *move that from physics for HS to somewhere else*
 - * *fix the algebra (T and not ΔT is appearing in the force...), checking against the main page [Small displacement:Statics:Thermodynamics](#)*

3.1.2 First principle for specific quantities

First principle and thermodynamics can be written in terms of specific quantities, usually either for unit volume or for unit mass.

First principle per unit mass

All the extensive quantities are written as the product of the mass of the system M and its density, namely

$$E = Me \quad , \quad S = Ms \quad , \quad \mathbf{X} = M\mathbf{x} \quad ,$$

and, using the product rule for differential $dE = d(Me) = dM e + M de$, first principle can be written as

$$0 = dM \underbrace{(e - Ts - \mathbf{F} \cdot \mathbf{x})}_{=0} + M (de - Tds - \mathbf{F} \cdot d\mathbf{x}) \quad .$$

The first term is identically zero, since it's the expression of the internal energy divided by M . Being $M > 0$, the second term gives the first principle per unit mass

$$de = Tds + \mathbf{F} \cdot d\mathbf{x} \quad .$$

Different systems

- Single-component fluid;

$$V = M \frac{1}{\rho} \quad , \quad N = M \frac{1}{m} \quad ,$$

being ρ the mass density and m the mass of a particle (or mole, it depends on the description) of the medium; m is **constant**. First principle becomes

$$0 = dM \underbrace{\left(e - Ts + \frac{P}{\rho} - \frac{\mu}{m} \right)}_{=0} + M \left(de - Tds - \frac{P}{\rho^2} d\rho \right) \quad ,$$

and thus

$$de = Tds + \frac{P}{\rho^2} d\rho \quad .$$

- Multi-component fluid

$$V = M \frac{1}{\rho} \quad , \quad N_k = M_k \frac{1}{m_k} = M \frac{M_k}{M} \frac{1}{m_k} = M \frac{1}{m_k} \frac{\rho_k}{\rho} = M \frac{1}{m_k} w_k \quad ,$$

being ρ the mass density and m the mass of a particle (or mole, it depends on the description) of the k^{th} substance; m_k is **constant**. The first principle becomes

$$0 = dM \underbrace{\left(e - Ts + \frac{P}{\rho} - \frac{\mu_k}{m_k} w_k \right)}_{=0} + M \left[de - Tds - \frac{P}{\rho^2} d\rho - \frac{\mu_k}{m_k} dw_k \right] \quad ,$$

and thus

$$de = Tds + \frac{P}{\rho^2} d\rho + \frac{\mu_k}{m_k} dw_k \quad .$$

- Single-component solid

todo

First principle per unit volume

All the extensive quantities are written as the product of the mass of the system M and its density, namely

$$E = V\mathcal{E} \quad , \quad S = M\mathcal{S} \quad , \quad \mathbf{X} = M\boldsymbol{\chi} \quad ,$$

and, using the product rule for differential $dE = d(V\mathcal{E}) = dV\mathcal{E} + Vd\mathcal{E}$, first principle can be written as

$$0 = dV \underbrace{(\mathcal{E} - T\mathcal{S} - \mathbf{F} \cdot \boldsymbol{\chi})}_{=0} + V(d\mathcal{E} - Td\mathcal{S} - \mathbf{F} \cdot d\boldsymbol{\chi}) \quad .$$

The first term is identically zero, since it's the expression of the internal energy divided by V . Being $V > 0$, the second term gives the first principle per unit volume

$$d\mathcal{E} = Td\mathcal{S} + \mathbf{F} \cdot d\boldsymbol{\chi} \quad .$$

Different systems

- Single-component fluid;

$$V = V \cdot 1 \quad , \quad N = V \frac{M}{V} \frac{1}{m} = V \frac{\rho}{m} \quad ,$$

being ρ the mass density and m the mass of a particle (or mole, it depends on the description) of the medium; m is **constant**. First principle becomes

$$0 = dV \underbrace{\left(\mathcal{E} - T\mathcal{S} + P - \mu \frac{\rho}{m}\right)}_{=0} + V \left(d\mathcal{E} - Td\mathcal{S} + 0 - \frac{\mu}{m} d\rho\right) \quad ,$$

and thus

$$d\mathcal{E} = Td\mathcal{S} + \frac{\mu}{m} d\rho \quad .$$

This latter formulation is consistent with the principle per unit mass. Volume density can be written as the product of mass density and the mass density of the physical quantity of interest, namely $\mathcal{E} = \rho e$

$$\begin{aligned} 0 &= -d(\rho e) + Td(\rho s) + \frac{\mu}{m} d\rho = \\ &= d\rho \left(-e + Ts + \frac{\mu}{m}\right) - \rho(de + Tds) = \\ &= d\rho \underbrace{\left(-e - \frac{P}{\rho} + Ts + \frac{\mu}{m}\right)}_{\text{def. of } e} + \rho \underbrace{\left(-de + \frac{P}{\rho^2} d\rho + Tds\right)}_{1^{st} \text{ pr. per unit mass}} \quad . \end{aligned}$$

- Multi-component fluid

todo

- Single-component solid

todo

3.2 Potentials - specific quantities

Here mechanical $\mathbf{X}_m$ and non-mechanical $\mathbf{X}_n$ generalized forces and displacements are recognized to define enthalpy and Gibbs' free energy later.

$$e(s, \mathbf{x}) = e\left(\frac{S}{M}, \frac{\mathbf{X}}{M}\right) = \frac{1}{M} E\left(\frac{S}{M}, \frac{\mathbf{X}}{M}\right) \quad \text{todo check}$$

Internal energy, $e(s, \mathbf{x})$.

$$de = Tds + \mathbf{F} \cdot d\mathbf{x}$$

Helmholtz free energy, $f(T, \mathbf{x}) = e - Ts$.

$$\begin{aligned} df &= de - Tds - sdT = \\ &= -sdT + \mathbf{F} \cdot d\mathbf{x}. \end{aligned}$$

Enthalpy, $h(T, \mathbf{F}_m, \mathbf{x}_n) = e - \mathbf{F}_m \cdot \mathbf{x}_m$.

$$\begin{aligned} dh &= de - d\mathbf{F}_m \cdot \mathbf{x}_m - \mathbf{F}_m \cdot d\mathbf{x}_m = \\ &= Tds - \mathbf{x}_m \cdot d\mathbf{F}_m + \mathbf{F}_n \cdot d\mathbf{x}_n. \end{aligned}$$

Gibbs' free energy, $g(T, \mathbf{F}_m, \mathbf{x}_n) = h - Ts = e - \mathbf{F}_m \cdot \mathbf{x}_m - Ts = f - \mathbf{F}_m \cdot \mathbf{x}_m = \mathbf{F}_n \cdot \mathbf{x}_n$.

$$\begin{aligned} dg &= dh - Tds - sdT = \\ &= -sdT - \mathbf{x}_m \cdot d\mathbf{F}_m + \mathbf{F}_n \cdot d\mathbf{x}_n = \\ &= \mathbf{x}_n \cdot d\mathbf{F}_n + \mathbf{F}_n \cdot d\mathbf{x}_n. \end{aligned}$$

Partial derivatives of potentials.

$$\begin{aligned} T &= \left. \frac{\partial e}{\partial s} \right|_{\mathbf{x}} = \left. \frac{\partial h}{\partial s} \right|_{\mathbf{F}_m, \mathbf{x}_n} \\ s &= - \left. \frac{\partial f}{\partial T} \right|_{\mathbf{x}} = - \left. \frac{\partial g}{\partial T} \right|_{\mathbf{F}_m, \mathbf{x}_n} \\ \mathbf{F}_m &= \left. \frac{\partial e}{\partial \mathbf{x}_m} \right|_{s, \mathbf{x}_n} = \left. \frac{\partial f}{\partial \mathbf{x}_m} \right|_{T, \mathbf{x}_n} \\ \mathbf{F}_n &= \left. \frac{\partial e}{\partial \mathbf{x}_n} \right|_{s, \mathbf{x}_m} = \left. \frac{\partial f}{\partial \mathbf{x}_n} \right|_{T, \mathbf{x}_m} = \left. \frac{\partial h}{\partial \mathbf{x}_n} \right|_{s, \mathbf{F}_m} = \left. \frac{\partial g}{\partial \mathbf{x}_n} \right|_{T, \mathbf{F}_m} \\ \mathbf{x}_m &= - \left. \frac{\partial h}{\partial \mathbf{F}_m} \right|_{s, \mathbf{x}_n} = - \left. \frac{\partial g}{\partial \mathbf{F}_m} \right|_{T, \mathbf{x}_n} \end{aligned}$$

Maxwell's relations.

todo Uncomment

THERMODYNAMIC COEFFICIENTS

In this section different related to first derivatives of thermodynamic state variables are introduced and discussed for different systems.

4.1 Thermodynamic coefficients of a single-component fluid

For a 1-component fluid with constant composition, the first principle reads

$$de = Tds - Pdv = Tds + \frac{P}{\rho^2}d\rho .$$

Heat capacity.

$$c_x := T \left. \frac{\partial s}{\partial T} \right|_x$$

For fluid systems, usually heat capacity at constant pressure or constant density are the most used.

Thermal expansion coefficients.

$$\alpha_x := \left. \frac{1}{v} \frac{\partial v}{\partial T} \right|_x = - \left. \frac{1}{\rho} \frac{\partial \rho}{\partial T} \right|_x$$

For fluid systems, usually thermal expansion coefficient at constant pressure is the most used.

Compressibility coefficients.

$$\beta_x := - \left. \frac{1}{v} \frac{\partial v}{\partial P} \right|_x = \left. \frac{1}{\rho} \frac{\partial \rho}{\partial P} \right|_x$$

For fluid systems, usually compressibility coefficient at constant temperature or entropy are the most used.

4.2 Relations between thermodynamic coefficients

Relation between c_v and c_p .

$$c_P - c_v = T v \frac{\alpha_P^2}{\beta_T} . \quad (4.1)$$

Relation between β_s and β_T - 1

$$\beta_s = \frac{c_v}{c_P} \beta_T . \quad (4.2)$$

Relation between β_s and β_T - 2

$$\beta_s - \beta_T = -\frac{vT}{c_P} \alpha_P^2. \quad (4.3)$$

Relation between β_s and β_T - 3

$$\frac{1}{\beta_s} - \frac{1}{\beta_T} = \frac{T}{vc_v} (\partial_T P|_v)^2. \quad (4.4)$$

Proof of the relation between heat capacities c_v, c_P

Changing independent variables from (T, v) to (P, v) in the expression of entropy $s(T, v) = s(T, P(v, T))$,

$$\begin{aligned} \frac{c_v}{T} &:= \left. \frac{\partial s}{\partial T} \right|_v = \left. \frac{\partial s}{\partial T} \right|_P + \left. \frac{\partial s}{\partial P} \right|_T \left. \frac{\partial P}{\partial T} \right|_v = & \text{(Maxwell } \partial_P s|_T = -\partial_T v|_P) \\ &= \left. \frac{\partial s}{\partial T} \right|_P - \left. \frac{\partial v}{\partial T} \right|_P \left. \frac{\partial P}{\partial T} \right|_v = & \text{(relation below } \partial_T P|_v = \dots) \\ &= \partial_T s|_P + \frac{(\partial_P v|_T)^2}{\partial_P v|_T} = & \text{(def of TD coeffs)} \\ &= \frac{c_P}{T} - \frac{v\alpha_P^2}{\beta_T} \end{aligned}$$

having exploited the relation

$$\left. \frac{\partial P}{\partial T} \right|_v = \frac{\partial(P, v)}{\partial(T, v)} = \frac{\partial(P, v)/\partial(T, P)}{\partial(T, v)/\partial(T, P)} = -\frac{\partial_P v|_T}{\partial_P v|_T}.$$

Proof of the relation between β_s and β_T - 1

$$\begin{aligned} v\beta_T &= \partial_P v|_T \\ v\beta_s &= \partial_P v|_s = \frac{\partial(v, s)}{\partial(P, s)} \frac{\partial(v, T)}{\partial(v, T)} \frac{\partial(P, T)}{\partial(P, T)} = \frac{\partial(P, T)}{\partial(P, s)} \frac{\partial(v, T)}{\partial(P, T)} \frac{\partial(v, s)}{\partial(v, T)} = \frac{c_v}{c_P} (v\beta_T) \\ &= \frac{1}{\partial_T s|_P} \end{aligned}$$

Proof of the relation between β_s and β_T - 2

$$\beta_s = \frac{c_v}{c_P} \beta_T = \frac{\beta_T}{c_P} \left[c_P - Tv \frac{\alpha_P^2}{\beta_T} \right] = \beta_T - Tv \frac{\alpha_P^2}{c_P}.$$

Proof of the relation between β_s and β_T - 3

$$\begin{aligned} \frac{1}{\beta_s} &= \frac{c_P}{c_v} \frac{1}{\beta_T} = \frac{1}{c_v \beta_T} \left[c_v + Tv \frac{\alpha_P^2}{\beta_T} \right] = \\ &= \frac{1}{\beta_T} \frac{1}{c_v} \left[c_v + Tv \frac{\alpha_P^2}{\beta_T} \right] = \\ &= \frac{1}{\beta_T} + Tv \frac{\frac{1}{v^2} (\partial_T v|_P)^2}{-\frac{1}{v} \partial_P v|_T} \frac{1}{-\frac{1}{v} \partial_P v|_T} = \\ &= \frac{1}{\beta_T} + Tv \left(\frac{\partial_T v|_P}{\partial_P v|_T} \right)^2 = \frac{1}{\beta_T} + Tv \left(\frac{\partial(v, P)}{\partial(T, P)} \frac{\partial(P, T)}{\partial(v, T)} \right)^2 = \frac{1}{\beta_T} + Tv \left(\left. \frac{\partial P}{\partial T} \right|_v \right)^2 \end{aligned}$$

4.3 Thermodynamic equilibrium

Here thermodynamic equilibrium is discussed for a single-component fluid, for which the first principle reads

$$de = Tds - Pdv = Tds + \frac{P}{\rho^2}d\rho.$$

First conditions on energy, $e(s, \rho)$, as a function of entropy and density then the equivalent conditions on entropy $s(e, \rho)$, as a function of energy and density are discussed.

4.3.1 Conditions on energy

Thermodynamic equilibrium implies conditions on the second order term in series expansion of the function $e(s, \rho)$ (**todo** why? Spend few words. Use Landau as a reference if needed)

$$\begin{aligned} e(s + ds, \rho + d\rho) &= e(s, \rho) + \partial_s e|_{\rho}(s, \rho)ds + \partial_{\rho} e|_s(s, \rho)d\rho + \\ &+ \frac{1}{2} [\partial_{ss} e|_{\rho}(s, \rho)ds^2 + 2\partial_{s\rho} e(s, \rho)dsd\rho + \partial_{\rho\rho} e(s, \rho)d\rho^2] . \end{aligned}$$

The equilibrium energy must be a *minimum* (**todo** check the function that must be a *minimum*, since $\partial_s e|_{\rho} = T, \dots$ and T not zero, and thus it can be a minimum!): the second order term must be positive for any increment of independent physical variables $d\rho, ds$, i.e.

$$\frac{1}{2} [\partial_{ss} e|_{\rho}ds^2 + 2\partial_{s\rho} e dsd\rho + \partial_{\rho\rho} e|_s d\rho^2]$$

is a positive-definite quadratic form, i.e. the Hessian

$$\begin{bmatrix} \partial_{ss} e|_{\rho} & \partial_{s\rho} e \\ \partial_{s\rho} e & \partial_{\rho\rho} e|_s \end{bmatrix}$$

is definite positive, and thus the following conditions must hold

$$\begin{cases} \partial_{ss} e|_{\rho} > 0 \\ \partial_{ss} e|_{\rho} \partial_{\rho\rho} e|_s - (\partial_{s\rho} e)^2 < 0 \end{cases}$$

4.3.2 Conditions on entropy

Using e, ρ as independent thermodynamic state variables, and the entropy $s(e, \rho)$ it can be proved that the condition on $e(s, \rho)$ for the thermodynamic equilibrium implies that $s(e, \rho)$ is a *maximum* (**todo** check the meaning of maximum here), and thus

$$\frac{1}{2} [\partial_{ee} s|_{\rho}de^2 + 2\partial_{e\rho} s ded\rho + \partial_{\rho\rho} s|_e d\rho^2]$$

is a positive-definite quadratic form, i.e. the Hessian

$$\begin{bmatrix} \partial_{ee} s|_{\rho} & \partial_{e\rho} s \\ \partial_{e\rho} s & \partial_{\rho\rho} s|_e \end{bmatrix}$$

is definite positive, and thus the following conditions must hold

$$\begin{cases} \partial_{ee} s|_{\rho} < 0 \\ \partial_{ee} s|_{\rho} \partial_{\rho\rho} s|_e - (\partial_{e\rho} s)^2 < 0 \end{cases}$$

Relation between partial derivatives of $e(s, \rho)$ and $s(e, \rho)$

The relation

$$e = e_{s\rho}(s_{e\rho}(e, \rho), \rho),$$

provides the link between the two representations, having written $e_{s,\rho}()$, $s_{e,\rho}()$ the functions with the argument indicated as indices. This relation contains only e, ρ as independent variables. All the required relations are evaluated computing partial derivatives of this relation.

First-order derivatives. It can be proved that

$$\begin{aligned}\partial_e s|_\rho &= \frac{1}{\partial_s e|_\rho} \\ \partial_\rho s|_e &= -\frac{\partial_\rho e|_s}{\partial_s e|_\rho}\end{aligned}$$

Second-order derivatives. It can be proved that

$$\begin{aligned}\partial_{ee} s|_\rho &= -\frac{1}{(\partial_s e|_\rho)^3} \partial_{ss} e|_\rho \\ \partial_{e\rho} s &= \frac{\partial_\rho e|_s}{(\partial_s e|_\rho)^3} \partial_{ss} e|_\rho - \frac{\partial_{s\rho} e}{(\partial_s e|_\rho)^2} \\ \partial_{\rho\rho} s|_e &= -\frac{1}{(\partial_s e|_\rho)^3} \left[\left(-\frac{\partial_\rho e|_s}{\partial_s e|_\rho} \right)^2 \partial_{ss} e|_\rho + 2 \left(-\frac{\partial_\rho e|_s}{\partial_s e|_\rho} \right) \partial_{s\rho} e + \partial_{\rho\rho} e|_s \right]\end{aligned}$$

Equivalence of conditions on $e(s, \rho)$ and on $s(e, \rho)$ for thermodynamic equilibrium

Exploiting first condition on partial derivatives of $e(s, \rho)$, the first condition on the partial derivatives of $s(e, \rho)$ is

$$\partial_{ee} s|_\rho = -\frac{1}{(\partial_s e|_\rho)^3} \partial_{ss} e|_\rho = -\frac{1}{T^3} \partial_{ss} e|_\rho < 0$$

since $T > 0$ and $\partial_{ss} e|_\rho > 0$.

Second condition for derivatives of $s(e, \rho)$ is

$$\begin{aligned}\partial_{\rho\rho} s|_e \partial_{ee} s|_\rho - (\partial_{\rho s} e)^2 &= \dots \\ &= \frac{1}{T^4} \left[\partial_{ss} e|_\rho \partial_{\rho\rho} e|_s - (\partial_{s\rho} e)^2 \right] < 0.\end{aligned}$$

STATI DELLA MATERIA E MODELLI

Stati della materia.

- gas
- liquidi
- solidi
- plasma

Alcune leggi costitutive.

- solidi elastici:
 - solidi lineari elastici isotropi
- fluidi:
 - in base all'equazione di stato:
 - * gas perfetti
 - * gas reali
 - * ...
 - in base all'espressione degli sforzi:
 - * fluidi newtoniani
 - * fluidi non-newtoniani

basics

Jul 13, 2026

0 min read

5.1 Gas ideali

Legge di Boyle-Mariotte. $PV = \text{cost.}$ a T costante (trasformazione isoterma).

Legge di Gay-Lussac I (o Charles). $V \propto T$ a P costante (trasformazione isobara).

Legge di Gay-Lussac II. $P \propto T$ a V costante (trasformazione isocora).

Legge dei gas ideali. $PV = nRT$

Part II

Thermal Engineering

PRINCIPLES OF THERMAL ENGINEERING

Thermal engineering is a discipline dealing with applications and systems for energy conversion and transfer through work and heat, usually involving other branches or field in physics/engineering, like thermodynamics, fluid mechanics, heat transfer, mass transfer, chemistry,...

Physical processes: continuum (usually fluid) mechanics, mass and heat transfer, chemistry,...

Applications: power plants, heat engines (direct cycles); refrigerator systems (inverse cycles); heat exchangers; heating; heating, ventilation, and air conditioning (HVAC); thermal insulation,...

Thermodynamic transformations and cycles

Heat transfer

THERMODYNAMIC TRANSFORMATIONS AND THERMOMECHANICAL SYSTEMS

7.1 Turbine

Integral balance equations for the operating fluid in the inner volume v_t of a turbine read

$$\begin{cases} \frac{d}{dt} \int_{v_t} \rho + \oint_{\partial v_t} \rho(\vec{u} - \vec{u}_b) \cdot \hat{n} = 0 \\ \frac{d}{dt} \int_{v_t} \rho \vec{u} + \oint_{\partial v_t} \rho \vec{u}(\vec{u} - \vec{u}_b) \cdot \hat{n} = \int_{v_t} \rho \vec{g} + \oint_{\partial v_t} \vec{t}_{\hat{n}} \\ \frac{d}{dt} \int_{v_t} \rho \vec{r} \times \vec{u} + \dots \\ \frac{d}{dt} \int_{v_t} \rho e^t + \oint_{\partial v_t} \rho e^t(\vec{u} - \vec{u}_b) \cdot \hat{n} = \int_{v_t} \rho \vec{g} \cdot \vec{u} + \oint_{\partial v_t} \vec{t}_{\hat{n}} \cdot \vec{u} - \oint_{\partial v_t} \vec{q} \cdot \hat{n} + \int_{v_t} \rho r. \end{cases}$$

For a Newtonian fluid, stress vector can be written as the sum of a pressure and viscous contribution,

$$\vec{t}_{\hat{n}} = -p\hat{n} + \vec{s}_{\hat{n}}.$$

The boundary of the geometrical domain ∂v_t can be represented as the union of the intake manifold s_{in} , exhaust manifold s_{out} , the steady solid boundary s_{sta} (stator,...), and rotatin solid boundary s_{rot} (rotor), here neglecting fuel intake (it can be considered as well, nothing changes radically). There's no mass flux through solid walls, i.e. $(\vec{u} - \vec{u}_b) \cdot \hat{n}|_{s_{sta}, s_{rot}} = 0$.

As a first approximation, volume force is considered negligible, $\vec{g} = \vec{0}$, heat transfer is considered negligible as well, $\vec{q} = \vec{0}$, $r = 0$ (and thus material particles undergo adiabatic transformations), and viscous stress is considered negligible at inflow and outflow, $\vec{s}_{\hat{n}}|_{s_{in}, s_{out}} = \vec{0}$, or $\vec{t}_{\hat{n}}|_{s_{in}, s_{out}} = -p\hat{n}|_{s_{in}, s_{out}}$. With all these assumptions, and further assuming steady regime $\frac{d}{dt} \equiv 0$, mass and total energy balance equations become

$$\begin{cases} \int_{s_{in}} \rho(\vec{v} - \vec{v}_b) \cdot \hat{n} + \int_{s_{out}} \rho(\vec{v} - \vec{v}_b) \cdot \hat{n} = 0 \\ \int_{s_{in}} \rho e^t(\vec{u} - \vec{u}_b) \cdot \hat{n} + \int_{s_{out}} \rho e^t(\vec{u} - \vec{u}_b) \cdot \hat{n} = - \int_{s_{in}} p\hat{n} \cdot \vec{v} - \int_{s_{out}} p\hat{n} \cdot \vec{v} + \int_{s_{rot}} \vec{t}_{\hat{n}} \cdot \vec{v}. \end{cases}$$

Mass balance equation reduces to equality of inflow and outflow mass flux: mass is not created or destroyed in classical mechanics, and can't accumulate in the volume in steady conditions. If $\vec{v}_b \cdot \hat{n} = 0$ (or other conditions to make the following manipulation of the total energy equation), total energy equation can be written in terms of fluxes of total enthalpy $h^t := e^t + \frac{P}{\rho}$,

$$\int_{s_{in}} \rho h^t \vec{u} \cdot \hat{n} + \int_{s_{out}} \rho e^t(\vec{u} - \vec{u}_b) \cdot \hat{n} = \int_{s_{rot}} \vec{t}_{\hat{n}} \cdot \vec{v},$$

i.e. the **total power done on the fluid by the turbine**¹, $P_{sf} = \int_{s_{rot}} \vec{t}_{\hat{n}} \cdot \vec{u}$, equals the difference of total entaphy fluxes through the inflow and outflow manifolds.

For action/reaction principle, 3rd principle of Newton mechanics, the stress transferred by the fluid to the solid walls of the turbine is $\vec{t}_{\hat{n},fs} = -\vec{t}_{\hat{n},sf}$; for boundary conditions (no slip for viscous fluids), the velocity of solid and fluid particle at walls concides, $\vec{v}_s = \vec{v}_f$, and thus the power transferred by the fluid to the solid reads

$$P_{fs} = \int_{s_{rot}} \vec{t}_{\hat{n},fs} \cdot \vec{v}_s = - \int_{s_{rot}} \vec{t}_{\hat{n},sf} \cdot \vec{v}_f = -P_{sf} .$$

Mass and total energy balance equations can be recast as

$$\begin{cases} \dot{m}_{in} = \dot{m}_{out} = \dot{m} \quad \text{const.} \\ \Phi(h^t)_{in} - \Phi(h^t)_{out} = P_{fs} . \end{cases}$$

being

$$\begin{aligned} \dot{m}_k &= \Phi(\rho)_k = - \int_{s_k} \rho \vec{v}_{rel} \cdot \hat{x} \\ \Phi(h^t)_k &= - \int_{s_k} \rho h^t \vec{v}_{rel} \cdot \hat{x} \end{aligned}$$

the mass and total enthalpy fluxes through the surfaces “in the main direction of the fluid”.

7.2 Compressor

¹ This contribution is the total power done by the turbine on the fluid, being the integral over the moving interface between the media of the dot product of the velocity of the material points and the stress acting *on the fluid*. Total power transferred from the turbine to the fluid could be written as the integral over the whole interface surface, namely $s_{sta} \cup s_{rot}$, but the power contribution of the stator is identically zero, being $\vec{v}|_{s_{sta}} = \vec{0}$.

THERMODYNAMIC CYCLES AND HEAT ENGINES

Direct and reversed thermodynamic cycles. Direct cycles: transform heat to mechanical work. Reversed cycle: transform mechanical work to heat transfer (e.g. refrigeration/cooling - refrigerator cycles - or heating - heat pump cycles)

Carnot cycle - reversible cycle between two constant temperature sources. Planck and Kelvin statements of 2nd principle of thermodynamics. Maximum efficiency....

Ideal models of real cycles. Otto, Diesel, Atkinson, Stirling (ICE), Rankine, Joule-Brayton

Principle. Reciprocating (piston)/reactive (turbine) engines

Circuit. Open/close

8.1 Carnot cycle and heat engine

8.1.1 Second principle of thermodynamics: Planck and Kelvin statements

8.2 Heat engines and cycles

In this section, the idealization/model of real thermodynamic cycles performed by real-life machines are discussed.

8.2.1 Otto

Introduction.

Applications.

Thermodynamic cycle. Reciprocating operation, with (approximately¹) close system between intake and exhaust processes:

- 0 → 1 intake, approximately isobaric
- 1 → 2 adiabatic compression
- 2 → 3 combustion, approximately isochoric at TDC: **spark ignition** usually makes combustion fast enough to be modeled as an immediate process occurring at TDC; this is in contrast to *Diesel engines*, where combustion starts with self-ignition due to high temperature reached at the end of the compression
- 3 → 4 adiabatic expansion
- 4 → 0 exhaust, approximately isochoric BDC followed by isobaric

¹ Combustion chamber is an approximately close system during a cycle between intake and exhaust, because of: 1. fuel injection, 2. leakage through the clearance between the walls of the piston and the cylinder.

8.2.2 Diesel

Introduction.

Applications.

Thermodynamic cycle. Reciprocating operation, with (approximately^{Page 37, 1}) close system between intake and exhaust processes:

- $0 \rightarrow 1$ intake, approximately isobaric
- $1 \rightarrow 2$ adiabatic compression
- $2 \rightarrow 3$ combustion, approximately isobaric at TDC: **self ignition** usually triggers slower combustion if compared with *Otto engines*, that can be modeled as an isobaric transformation
- $3 \rightarrow 4$ adiabatic expansion
- $4 \rightarrow 0$ exhaust, approximately isochoric BDC followed by isobaric

8.2.3 Joule-Brayton

Introduction. It's the reference thermodynamic cycle for **gas turbine** heat engines. These systems have continuous operation and can work both in open (jet propulsion) and close cycles (electric power generation).

Applications. Joule-Brayton close cycle is used in electric power generation: heat produced by combustion is transferred to an operating fluid (chemical power to thermal power), entering a turbine that converts "thermal power of the fluid" into mechanical power, then converted to electric power by an electric generator. Joule-Brayton open cycle is used for jet propulsion (typically for aircraft): operating fluid is usually air, undergoing the thermodynamic cycle through the jet engine; air is compressed before entering the combustion chamber, and released after passing through a turbine: turbine extracts the power required to drive the compressor, and "extra power" contained in the fluid is used to accelerating exhaust fluid, producing thrust.

Thermodynamic cycle - close cycle. Components involved in the system are usually open systems. Steady operation of these systems is described by balance of total enthalpy flux. Kinetic energy of the fluid is usually negligible if compared with internal energy and enthalpy: total enthalpy reduces to **enthalpy**

- $1 \rightarrow 2$ adiabatic compression
- $2 \rightarrow 3$ combustion, approximately isobaric at in combustion chamber
- $3 \rightarrow 4$ adiabatic expansion
- $4 \rightarrow 1$ isobaric cooling

Thermodynamic cycle - open cycle. Components involved in the system are usually open systems. Steady operation of these systems is described by balance of **total enthalpy** flux. Kinetic energy of the fluid is usually **not** negligible if compared with internal energy and enthalpy

- $0 \rightarrow 1$ adiabatic compression in free air before or at the engine intake
- $1 \rightarrow 2$ adiabatic compression in compressor
- $2 \rightarrow 3$ combustion, approximately isobaric at in combustion chamber
- $3 \rightarrow 4$ adiabatic expansion in turbine
- $4 \rightarrow 5$ adiabatic expansion in the nozzle

8.2.4 Rankine

Introduction.

Application.

Thermodynamic cycle.

8.3 Applications

8.3.1 Mechanical power

8.3.2 Power generation

8.3.3 Heat pump and/or refrigeration

HEAT TRANSFER

Three main heat transfer mechanisms exist: conduction, convection, radiation.

Conduction. Main heat transfer mechanism in solid media: molecules of the solid are not free to move and conduction is a diffusion process of the microscopical thermal agitation of the molecules of the solid structure.

Convection. Main heat transfer mechanism in fluid media: molecules of the fluid are free to move; as they move, they transport “thermal energy” and exchange heat with the surrounding environment.

Radiation. It’s the only heat transfer mechanism that needs no matter to occur, as it can occur in “vacuum”¹

9.1 Conduction

$$\rho c \frac{\partial T}{\partial t} = -\nabla \cdot \vec{q} + \rho r$$

Fourier model of heat conduction flux wants the heat flux to be proportional to the temperature gradient, ∇T . The most general expression for linear non-isotropic media reads

$$\vec{q} = -\mathbb{K} \cdot \nabla T ,$$

with $\mathbb{K}$ the conductivity tensor (a second-order (semi)definite (symmetric? **todo check, Onsager reciprocal relations**) positive to comply with the second principle of thermodynamics: “heat transfers energy from hot to cold regions”). The behavior of isotropic media is described by isotropic tensors; thus¹ the conductivity tensor becomes $\mathbb{K}^{iso} = k\mathbb{I}$, and the heat flux reads

$$\vec{q} = -k \nabla T .$$

Differential problem. It follows that conduction in a ... medium in region Ω for time $t \in [t_0, t_1]$ is governed by the differential problem represented by the PDE

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \rho r \quad \vec{r} \in \Omega ,$$

supplied with proper initial conditions in every point $\vec{r} \in \Omega$

$$T(\vec{r}, t_0) = T_0(\vec{r}) \quad \vec{r} \in \Omega$$

and boundary conditions for every point on the boundary of the domain, $\vec{r} \in \partial\Omega$ for every time $t \in [t_0, t_1]$. Different boundary conditions may represent different physical processes:

¹ Vacuum of matter, mass, but not physical properties, as the dielectric constant and magnetic permeability of free space/“vacuum”.

¹ Isotropic 2-nd order tensors are the identity tensor and its multiples, $\mathbb{A}^{iso} = a\mathbb{I}$. The dot product of 2-nd order identity tensor and a vector (field) gives the vector itself, $\mathbb{I} \cdot \vec{v} = \vec{v}$.

- known temperature on S_D (Dirichlet boundary conditions: the value of the unknown function is prescribed),

$$T(\vec{r}, t) = T_D(\vec{r}, t) \quad \vec{r} \in S_D$$

- known heat flux on S_N (Neumann boundary conditions: the value of the directional derivative of the unknown function is prescribed, in the normal direction w.r.t. the boundary),

$$\hat{n}(\vec{r}, t) \cdot \nabla T(\vec{r}, t) = \phi_N(\vec{r}, t) \quad \vec{r} \in S_N$$

- known value of a linear combination of the value of the unknown function and its directional derivative (Robin boundary conditions),

$$\alpha T(\vec{r}, t) + \beta \hat{n}(\vec{r}, t) \cdot \nabla T(\vec{r}, t) = h(\vec{r}, t) \quad \vec{r} \in S_R$$

todo

- derivation of this equation from governing equations of continuum mechanics: list assumptions, thermodynamic conditions (to transform internal energy e as a function of T)
 - link to *Mathematics:Numerical methods for PDEs:Elliptic (steady) and Parabolic (unsteady) problems*
-

9.2 Convection

9.3 Radiation

check all the nomenclature/definition

9.3.1 Planck's law

Definition 9.3.1 (Black-body)

Spectral energy density for black-body radiation

$$u_f(f, T) = \frac{8\pi h f^3}{c^3} \frac{1}{e^{\frac{hf}{k_B T}} - 1}$$

Spectral energy density has the physical dimension $\frac{\text{Energy}}{\text{Volume} \times \text{Frequency}}$

Spectral radiance reads¹ $B_f = \frac{1}{4\pi} u_f(f, T) c$ and thus for a black-body

$$B_f(f, T) = \frac{2hf^3}{c^2} \frac{1}{e^{\frac{hf}{k_B T}} - 1}.$$

It has physical dimension $\frac{\text{Energy}}{\text{Volume} \times \text{Frequency}} \times \text{Velocity} = \frac{\text{Power}}{\text{Area} \times \text{Frequency}}.$

¹ Show how to get this expression from the spectral energy density.

9.3.2 Wien's law

Wien's law states that the maximum of the spectral energy is obtained at a frequency f^* , whose value is proportional with the temperature of the body.

$$f^* = x^* \frac{k_B}{h} T \simeq 2.82 \frac{k_B}{h} T ,$$

where $x^* \simeq 2.82$ is the solution of the algebraic equation $3(1 - e^x) - x = 0$.

Proof of Wien's law

From direct evaluation of the derivative of the spectral radiance as a function of f ,

$$\begin{aligned} \partial_f B_f(f, T) &= \frac{2h}{c^2} \left[3f^2 \frac{1}{e^{\frac{hf}{k_B T}} - 1} + f^3 \left(-\frac{\frac{h}{k_B T} e^{\frac{hf}{k_B T}}}{\left(e^{\frac{hf}{k_B T}} - 1 \right)^2} \right) \right] = \\ &= \frac{2hf^2 e^{\frac{hf}{k_B T}}}{c^2 \left(e^{\frac{hf}{k_B T}} - 1 \right)^2} \left[3 \left(1 - e^{-\frac{hf}{k_B T}} \right) - \frac{hf}{k_B T} \right] . \end{aligned}$$

Now, if $\partial_f B_f(f, T) = 0$ the frequency is either $f = 0$, or the solution of the nonlinear algebraic equation

$$0 = 3 \left(1 - e^{-\frac{hf}{k_B T}} \right) - \frac{hf}{k_B T} .$$

Defining $x := \frac{hf}{k_B T}$, this equation becomes

$$0 = 3(1 - e^x) - x ,$$

whose solution $x^* \approx 2.82$ can be easily evaluated with an iterative method (or expressed in term of the Lambert's function W , so loved at Stanford and on Youtube: they'd probaly like to look at tabulated values, or pose). Once the solution x^* of this non-dimensional equation is found, the frequency where maximum energy density occurs reads

$$f^* = \frac{k_B T}{h} x^* \simeq 2.82 \frac{k_B}{h} T .$$

9.3.3 Energy transfer by radiation

A list of physical quantities used to describe a radiation process follows.

- **Radiant flux**, Φ_e . Physical dimension: W
- **Radiant intensity** $I_{e,\Omega}$ is the radiant flux per unit solid angle Ω . Physical dimension: $\frac{W}{sr}$
- **Spectral radiant intensity** $I_{e,\Omega,f}$ is the radiant flux per unit solid angle per frequency of the radiation: $\frac{W}{sr \text{ Hz}}$
- **Radiant exitance**, M_e is the flux across an emispherical receiver surface, per unit surface of the source. Physical dimension $\frac{W}{Area}$

Definition 9.3.2 (Lambert medium)

A Lambert medium is defined as a medium that follows **Lambert cosine law**,

$$I_1(\vec{r}_{12}, \hat{n}_1) = I_{1,0} \hat{r}_{12} \cdot \hat{n}_1 ,$$

with $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$.

Elementary power flux of the radiation emitted by a source in 1 through a solid angle $d\Omega_{21}$, as seen from 1, reads

$$dP_{1 \rightarrow 2} = I_1(\vec{r}_{12}, \hat{n}_1) d\Omega_{2/1}$$

The solid angle $d\Omega_{21}$ of a surface dS_2 as seen from a point in $\vec{r}_1$ is $d\Omega_{21} = \frac{\hat{n}_2 \cdot \vec{r}_{12}}{|\vec{r}_{12}|^3} dS_2$. From the expression of the elementary power flux,

$$dP_{1 \rightarrow 2} = I_1(\vec{r}_{12}, \hat{n}_1) \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3} \cdot \hat{n}_2 dS_2 = \vec{s}_{12} \cdot \hat{n}_2 dS_2 ,$$

it's possible to find the expression of the power flux density vector

$$\vec{s}_{12} = I_{12} \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3} .$$

For a Lambert medium then

$$\begin{aligned} dP_{1 \rightarrow 2} &= I_1(\vec{r}_{12}, \hat{n}_1) \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3} \cdot \hat{n}_2 dS_2 = \\ &= I_{0,1} \frac{\vec{r}_{12}}{|\vec{r}_{12}|} \cdot \hat{n}_1 \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3} \cdot \hat{n}_2 dS_2 . \end{aligned} \quad (9.1)$$

Stefan-Boltzmann law

The **radiant exitance**, M_e° , - i.e. the total power through an emisphere emitted by a source, per unit area of the source - of a black body at temperature T_1 is

$$M_e^\circ = \sigma T_1^4 ,$$

with the **Stefan-Boltzmann constant**, $\sigma = 5.67 \cdot 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}$.

Total radiation and Stefan-Boltzmann constant σ

Integration over frequency $f \in [0, +\infty]$, and over an emisphere receiver surface (or better, over the equivalent solid angle) for a Lambert medium provide the **radiant exitance**

$$\begin{aligned} M_e^\circ &= M_{1,2\text{emisphère}} = \frac{dP_{1 \rightarrow 2}}{dS_1} = \\ &= \int_{f=0}^{+\infty} \int_{\Omega_{\text{emisphère}}} I_{1,0}(f) \cos \theta_{12} df d\Omega_{2/1} = \\ &= \int_{f=0}^{+\infty} I_{1,0}(f) df \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi_2} \cos \theta \sin \theta d\theta d\phi , \end{aligned}$$

with the solid angle with spherical coordinates, $d\Omega_{2/1} = \sin \theta d\theta d\phi$. The geometric integral gives

$$\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi_2} \cos \theta d\theta d\phi = \pi .$$

For a black-body the integral over frequency gives

$$\begin{aligned} \int_{f=0}^{+\infty} I_{1,0}(f) df &= \int_{f=0}^{+\infty} \frac{2hf^3}{c^2} \frac{1}{e^{\frac{hf}{k_B T}} - 1} df \\ &= T^4 \frac{2h}{c^2} \left(\frac{k_B}{h} \right)^4 \int_{u=0}^{+\infty} \frac{u^3}{e^u - 1} du =: T^4 \frac{\sigma}{\pi} , \end{aligned}$$

i.e. it's proportional to T^4 with a constant of proportionality built on constants of nature, speed of light c , Boltzmann constant k_B , and Planck constant h . The integral is non-dimensional and its value² is $\frac{\pi^4}{15}$. **Stefan-Boltzmann constant** σ is then defined as

$$\sigma := \pi \cdot \frac{2h}{c^2} \left(\frac{k_B}{h} \right)^4 \int_{u=0}^{+\infty} \frac{u^3}{e^u - 1} du = \frac{2\pi^5}{15} \frac{k_B^4}{c^2 h^3} = 5.67 \cdot 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}.$$

Integration over a generic surface S_2 of the power flux emitted by a source 1 per unit surface of the source gives

$$M_{12} = \frac{dP_{1 \rightarrow 2}}{dS_1} = T_1^4 \frac{\sigma}{\pi} \int_{S_2} \hat{n}_1 \cdot \frac{\vec{r}_{12}}{|\vec{r}_{12}|} \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3} \cdot \hat{n}_2 dS_2$$

Thus the elementary power per unit surface of the source 1 and the receiver 2 is

$$dP_{1 \rightarrow 2} = T_1^4 \frac{\sigma}{\pi} \hat{n}_1 \cdot \frac{\vec{r}_{12}}{|\vec{r}_{12}|} \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3} \cdot \hat{n}_2 dS_1 dS_2$$

Comparing this expression of the power flux from source 1 to the elementary surface dS_2 with the expression (9.1) using the radiant intensity $I_{0,1}$, it immediately follows that the elementary radiant intensity $dI_{0,1}$ of the elementary surface dS_1 of the source reads

$$dI_{0,1} = T_1^4 \frac{\sigma}{\pi} dS_1.$$

Geometry effects on radiation

The elementary power is the product of $\frac{\sigma T^4}{\pi}$ and a factor depending only on the geometry of the problem,

$$dG_{12} := \hat{n}_1 \cdot \frac{\vec{r}_{12}}{|\vec{r}_{12}|} \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3} \cdot \hat{n}_2 dS_1 dS_2.$$

This factor is **symmetric**, i.e. inverting the roles of the source and the receiver is irrelevant. Letting $G_{12} = \int_{S_1} \int_{S_2} dG_{12}$, if the source surface 1 has uniform temperature T_1 and the receiver surface has uniform temperature T_2 , (and assuming *emissivity* $\varepsilon = 1$, for black-bodies) thus the power fluxes from 1 to 2 and from 2 to 1 read

$$\begin{aligned} P_{1 \rightarrow 2} &= T_1^4 \frac{\sigma}{\pi} G_{12} \\ P_{2 \rightarrow 1} &= T_2^4 \frac{\sigma}{\pi} G_{12}. \end{aligned}$$

Thus the net power into surfaces 1 and 2 are

$$\begin{aligned} P_{\rightarrow 1} &= P_{2 \rightarrow 1} - P_{1 \rightarrow 2} \\ P_{\rightarrow 2} &= P_{1 \rightarrow 2} - P_{2 \rightarrow 1}, \end{aligned}$$

so that $P_{\rightarrow 1} + P_{\rightarrow 2} = 0$.

In the limit of **small surfaces compared with their distance**, an approximate expression of the integral $\int_{S_1} \int_{S_2} dG$ allows to write the power transmitted by radiation from surface 1 to surface 2 as

$$P_{1 \rightarrow 2} = \frac{\sigma T_1^4}{\pi} G_{12} = \frac{\sigma T_1^4}{\pi} A_{1, \perp \vec{r}_{12}} \Omega_{2/1} = \frac{\sigma T_1^4}{\pi} A_{2, \perp \vec{r}_{12}} \Omega_{1/2}$$

i.e. as the product of the radiant ... **todo find the right term** $\frac{\sigma T_1^4}{\pi}$, the projection of the surface S_1 in a plane orthogonal to the distance vector $\vec{r}_{12}$ and the solid angle $\Omega_{2/1}$ of the surface 2 as seen by surface 1, or viceversa.

² **todo** Evaluate it

Proof of the small surface limit

Emissivity and coefficient of absorption

The radiation emission of a generic body differs from the radiation of a black-body. Its emittance can be written comparing its emission with that of a black body. In general, the emission could differ as a function of the frequency of the radiation and the direction of the radiation.

...

If there's no dependence on f and $\hat{r}$ - or taking the average values, neglecting the dependence on f and $\hat{r}$ if using the average coefficient - the radiant exitance is

$$M_1 = \varepsilon_1 \sigma T_1^4 .$$

Reflection coefficient r and absorption (or transmission) coefficient a are then defined as the reflected and absorbed ratios of the incident radiation. Given incident radiation with power P_i , power of reflected and absorbed radiation reads

$$P_r = r P_i \quad , \quad P_a = a P_i .$$

and for “power balance”, the sum of absorbed and reflected power must be equal to the incident power

$$P_i = P_r + P_a ,$$

and thus the coefficients must satisfy the following relation **check** *check the definition of the coefficients, if they should take into account the emission...*

$$1 = r + a .$$

Example 9.3.1 (Energy exchange in isolated systems and relations between coefficients - todo check)

Radiation transmission is assumed to be instantaneous here: this assumption holds for bodies close enough for the time $\Delta t = \frac{d_{12}}{c}$ to be small enough to be treated as zero.

A system composed of two bodies. Energy balance of a body immediately follows the first principle of thermodynamics: time derivative of energy of a system equals the net power done on the system. Here the net power equals the difference between the absorbed incoming power and the emitted power,

$$\begin{cases} \dot{E}_1 = P_{\rightarrow 1}^a - P_1^e \\ \dot{E}_2 = P_{\rightarrow 2}^a - P_2^e \end{cases}$$

Incoming power. The incoming power is due to the power emitted and reflected by the other body,

$$\begin{cases} P_{\rightarrow 1} = P_2^e + r_2 P_{\rightarrow 2} \\ P_{\rightarrow 2} = P_1^e + r_1 P_{\rightarrow 1} \end{cases}$$

From these two relations, the incoming powers can be explicitly written as function of the emitted powers,

$$\begin{cases} P_{\rightarrow 1} = \frac{P_2^e + r_2 P_1^e}{1 - r_1 r_2} \\ P_{\rightarrow 2} = \frac{P_1^e + r_1 P_2^e}{1 - r_1 r_2} \end{cases}$$

Energy balance of the system. Energy of an isolated system is constant and thus the time derivative of the energy is zero,

$$\begin{aligned}
 0 &= \dot{E} = \dot{E}_1 + \dot{E}_2 = \\
 &= a_1 \frac{P_2^e + r_2 P_1^e}{1 - r_1 r_2} - P_1^e + a_2 \frac{P_1^e + r_1 P_2^e}{1 - r_1 r_2} - P_2^e = \\
 &= P_1 \frac{a_2 + a_1 r_2 - 1 + r_1 r_2}{1 - r_1 r_2} + P_2 \frac{a_1 + a_2 r_1 - 1 + r_1 r_2}{1 - r_1 r_2} \\
 &= T_1^4 \sigma \varepsilon_1 \frac{a_2 + a_1 r_2 - 1 + r_1 r_2}{1 - r_1 r_2} + T_2^4 \sigma \varepsilon_2 \frac{a_1 + a_2 r_1 - 1 + r_1 r_2}{1 - r_1 r_2} .
 \end{aligned}$$

If this relation must hold in any condition, for any value of the emitted powers and for any independent choice of the materials of the bodies, the coefficients must be zero

$$\begin{cases} a_2 + a_1 r_2 - 1 + r_1 r_2 = 0 \\ a_1 + a_2 r_1 - 1 + r_1 r_2 = 0 \end{cases}$$

Subtracting,

$$a_2(1 - r_1) = a_1(1 - r_2) ,$$

and, since the relation must hold for any choice of pair of materials, the relation $a_1 = 1 - r_1$ is found again.

Bodies in thermal equilibrium. If each body is in thermal equilibrium,

$$\begin{cases} 0 = \frac{P_2^e + r_2 P_1^e}{1 - r_1 r_2} - P_1^e = \sigma G_{12} \frac{\varepsilon_2 T_2^4 + (r_2 - 1 + r_1 r_2) \varepsilon_1 T_1^4}{1 - r_1 r_2} \\ 0 = \frac{P_1^e + r_1 P_2^e}{1 - r_1 r_2} - P_2^e = \sigma G_{12} \frac{\varepsilon_1 T_1^4 + (r_1 - 1 + r_1 r_2) \varepsilon_2 T_2^4}{1 - r_1 r_2} \end{cases}$$

Kirchhoff's law

For systems in thermodynamic equilibrium,...

- constant ratio of ...
- $\alpha = \varepsilon$? for a system in thermal equilibrium **only**

Part III

Electro-chemical systems

PRINCIPLES OF THERMODYNAMICS - ELECTRO-CHEMICAL SYSTEMS

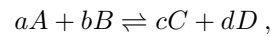
In this section:

- Introduction to *chemical reactions*
- *Energy balance*
- *Law of mass action*

In next sections:

- *Batteries*

10.1 Chemical reactions



or

$$\nu_{R_i} R_i \rightleftharpoons \nu_{P_j} P_j \tag{10.1}$$

10.2 First principle - energy balance

Energy balance

Total energy balance

$$dE^{tot} = \delta L^{ext} + \delta Q^{ext} .$$

Macroscopic kinetic energy balance (theorem, from mechanics)

$$dK = \delta L^{ext} + \delta L^{int} .$$

Internal energy $E := E^{tot} - K$

$$dE = -\delta L^{int} + \delta Q^{ext} .$$

Internal energy, intensive and extensive variables

$$dE = -\delta L^{int,rev} + \underbrace{\delta^+ D + \delta Q^{ext}}_{=: TdS}$$

As dE is a state function, and $\delta L^{int,rev}$ must be reversible by definition, the remaining terms must be reversible as well and completing the exact differential of the energy as a state function. The internal nature may have different natures, e.g. mechanical, electrical, magnetic, chemical

$$\delta L^{int,rev} = \delta L^{int,mech} + \delta L^{int,el} + \delta L^{int,magn} + \delta L^{int,el} + \dots$$

but all of these works can be written as the product of an intensive variable $\mathbf{F}$ of the system and the variation of an extensive variable $\mathbf{X}$, formally

$$\delta L^{int,rev} = \mathbf{F} \cdot d\mathbf{X},$$

- mechanical work, depends on the nature of the system. As an example, for a gas $\delta L^{int,rev} = -PdV$, or for an elastic solid $\delta L^{int,rev} = -V\sigma : d\varepsilon$
- electrical work, being V the (uniform) electrical potential of the system, and Q its electrical charge (see [Electrostatic energy: systems with uniform potential](#))

$$-VdQ$$

- chemical work, with μ_i the molar chemical potential i of the substance (the amount of internal energy added to the system by a change in the number of moles of the i substance in the system)

$$\mu_i dN_i$$

todo See and uniform the treatment in First principle, Second principle, Gibbs formalization, [Thermodynamic potentials](#)
Using $\mathbf{X}, S$ as independent variables,

$$dE = \mathbf{F}(\mathbf{X}, S) \cdot d\mathbf{X} + T(\mathbf{X}, S)dS.$$

If the energy of the system is a order-1 homogeneous function of the extensive variables of the system, $E(\lambda\mathbf{Y}) = \lambda E(\mathbf{Y})^1$, Euler theorem ([Theorem 3.1.1](#)) gives

$$E = TS + \mathbf{F} \cdot \mathbf{X}.$$

Explicitly writing the number of moles of the substances of the system out from the extensive variables $\mathbf{X}$, $\mathbf{X} \rightarrow (\mathbf{X}, \mathbf{N})$, and the chemical potentials out from the generalized intensive variables $\mathbf{F} \rightarrow (\mathbf{F}, \mu)$, the expression of the internal energy and its differential reads

$$E = TS + \mu \cdot \mathbf{N} + \mathbf{F} \cdot \mathbf{X}$$

$$dE = TdS + \mu \cdot d\mathbf{N} + \mathbf{F} \cdot d\mathbf{X}.$$

10.3 Law of Mass Action

The law of mass action states that the forward velocity v_f ($R \rightarrow P$) and backward velocity v_b ($R \leftarrow P$) of reaction (10.1) reads

$$v_f := -\frac{1}{\nu_{R_k}} \frac{d[R_k]_f}{dt} := \frac{1}{\nu_{P_k}} \frac{d[P_k]_f}{dt} = k_f \prod_i [R_i]^{\alpha_{R_i}}$$

$$v_r := \frac{1}{\nu_{R_k}} \frac{d[R_k]_r}{dt} := -\frac{1}{\nu_{P_k}} \frac{d[P_k]_r}{dt} = k_r \prod_i [P_i]^{\alpha_{P_i}}$$

¹ Not all the systems are made by sub-systems that combine “extensively”: it’s likely to be impossible to write the energy of these systems as a homogeneous function of the extensive variables of the system.

Taking $R \rightarrow P$ as the positive direction of the reaction, the velocity of the reaction is defined as $v = v_r - v_f$, and the rate of change of the concentrations thus reads

$$\begin{aligned}\frac{d[R_k]}{dt} &= \frac{d[R_k]_f}{dt} + \frac{d[R_k]_r}{dt} = \nu_{R_k} (v_r - v_f) = \nu_{R_k} v \\ \frac{d[P_k]}{dt} &= \frac{d[P_k]_f}{dt} + \frac{d[P_k]_r}{dt} = \nu_{P_k} (v_f - v_r) = \nu_{P_k} v.\end{aligned}$$

- **Equilibrium** is reached when $v_r = v_f$, and thus the rate of change of concentration of the substances is zero.
- **Extent of reaction** ξ , so that $d\xi = v dt$.

Constant k_r and k_f usually can be written as

$$k = A \exp\left(-\frac{E}{kT}\right),$$

being E the **activation energy**, k Boltzmann constant, T the temperature, and A a constant summarizing all the other conditions for the reaction to occur.

References

- https://en.wikipedia.org/wiki/Chemical_equilibrium
-

NERNST EQUATION

...

$$\mu_i = \mu_i^\circ - RT \ln a_i .$$

todo

- derivation:
 - with classical thermodynamics (for mixture of ideal gas, and then correct with the activity for the deviation from this behavior)
 - with statistical mechanics

BATTERIES: THERMO-ELECTRO-CHEMICAL MODELING

A battery is an electrochemical system that converts chemical energy directly into electrical energy via spontaneous reduction-oxidation (redox) reactions. It acts as an electrochemical reactor where the spatial separation of the oxidation and reduction steps forces electrons to flow through an external load to perform work.

Core components.

- Anode (negative electrode): the electrode where oxidation (loss of electrons) occurs during discharge.
- Cathode (positive electrode): the electrode where reduction (gain of electrons) occurs during discharge.
- Electrolyte: a medium containing mobile ions that carries ionic current inside the cell while remaining electronically insulating.
- Separator: a permeable membrane that prevents physical contact between the anode and cathode to avoid internal short circuits.

Classification.

- Primary (non-rechargeable): the underlying chemical reactions are thermodynamically irreversible or physically difficult to reverse. Once the reactants are depleted, the battery cannot be recharged (e.g., Alkaline, Zinc-carbon).
- Secondary (rechargeable): the redox reactions are highly reversible. Applying an external electrical potential drives the current in reverse, regenerating the initial chemical reactants (e.g., Lithium-ion, Lead-acid, NiMH).

Examples.

- *Daniell cell*
- *Li-ion battery*

12.1 Daniell cell

12.1.1 Description

The Daniell cell consists of two separate half-cells:

- A zinc (Zn) electrode immersed in a zinc sulfate (ZnSO_4) aqueous solution, with ions Zn^{2+} , SO_4^{2-} fully dissolved in that
- A copper (Cu) electrode immersed in a copper(II) sulfate (CuSO_4) aqueous solution, ions Cu^{2+} , SO_4^{2-} fully dissolved in that
- A porous barrier (or salt bridge) separating the solutions to prevent rapid mixing while allowing sulfate ions (SO_4^{2-}) to migrate, between the 2 half-cells.

The spontaneous half-cell reactions during discharge are:

- at the anode:



- at the cathode:



12.1.2 Regimes

- *Open circuit equilibrium*: chemical reactions at equilibrium, no net motion of charges through the electrolytes and the salt bridge
- *Operation under load*. Chemical kinetics reactions and diffusion process of ions may produce bottlenecks in the process, and thus overpotentials and/or losses

Open-circuit equilibrium

Internal energy

$$dE = -PdV + TdS + \sum_i \mu_i dN_i + \sum_k \phi_k dQ_k$$

with i spanning all the substances involved in a reaction, and k all the regions with uniform electric potential,

$$\begin{aligned} i &\in \{\text{Zn}, \text{Zn}^{2+}, \text{Cu}^{2+}, \text{Cu}\} \\ k &\in \{\text{anode}, \text{cathode}, \text{electrolytes+bridge}\} \end{aligned}$$

Here the chemical potentials include both the energy changes associated with reaction and phase transformation. Using the reactions (12.1), (12.2), it follows (with $z = 2$, number of electrons per reacting ion)

$$\begin{aligned} dQ_{\text{anode}} &= -zqdn_{\text{Zn}^{2+}} = -zqN_A dN_{\text{Zn}^{2+}} = -zFdN_{\text{Zn}^{2+}} \\ dQ_{\text{cathode}} &= -zqdn_{\text{Cu}^{2+}} = -zqN_A dN_{\text{Cu}^{2+}} = -zFdN_{\text{Cu}^{2+}}, \end{aligned}$$

having introduced the **Faraday's constant**, representing the electrical charge per mole of ions (with electrical charge q per particle),

$$F = qN_A = 96485.33 \frac{\text{C}}{\text{mol}},$$

defined as the product of the elementary charge (electron charge, but positive), and the number of Avogadro N_A .

If the motion of SO_4^{2-} ions through the salt bridge keeps the electrolytes electrically neutral, it's possible to write the variation of the moles of substances in the system as a function of an **extent of reaction variable**, ξ

$$dN_{\text{Zn}^{2+}} = dN_{\text{SO}_4^{2-},a} = -dN_{\text{SO}_4^{2-},c} = -dN_{\text{Cu}^{2+}} = d\xi.$$

If the electrolytes and the salt bridge are in an equi-potential region, then the motion of the SO_4^{2-} ions doesn't influence the variation of the internal energy, that can be written as

$$\begin{aligned} dE &= -PdV + TdS + (\mu_{\text{Zn}^{2+}} - \mu_{\text{Zn}} - \mu_{\text{Cu}^{2+}} + \mu_{\text{Cu}}) d\xi - zF\phi_a d\xi + zF\phi_c d\xi = \\ &= -PdV + TdS + [(\mu_{\text{Zn}^{2+}} - \mu_{\text{Zn}} - \mu_{\text{Cu}^{2+}} + \mu_{\text{Cu}}) - zF(\phi_a - \phi_c)] d\xi. \end{aligned}$$

Gibbs' free energy, $G = E + PV - TS$.

$$dG = VdP + SdT + \sum_i \mu_i dN_i - \sum_k \phi_k dQ_k$$

Equilibrium condition $dG = 0$ (see Thermodynamic equilibrium: Gibbs's free energy **todo**) for constant P and T (and thus $dP = 0$, $dT = 0$), requires that the electro-chemical force must be zero, i.e.

$$0 = \mu_{\text{Zn}^{2+}} - \mu_{\text{Zn}} - \mu_{\text{Cu}^{2+}} + \mu_{\text{Cu}} - zF(\phi_a - \phi_c) .$$

Thus, the electrostatic potential at open-circuit equilibrium between the cathode (positive electrode) and the anode (negative electrode) reads

$$V_{oc} = \phi_c - \phi_a = \frac{1}{zF} (-\mu_{\text{Zn}^{2+}} + \mu_{\text{Zn}} + \mu_{\text{Cu}^{2+}} - \mu_{\text{Cu}})$$

Introducing Nernst equation for the chemical potentials, the open-circuit potential can be written as

$$\begin{aligned} V_{oc} &= \frac{1}{zF} \sum_i \{ \text{sign}_i \mu_i + RT \ln a_i \} = \\ &= \frac{1}{zF} \underbrace{ \{ -\mu_{\text{Zn}^{2+}}^\circ + \mu_{\text{Zn}}^\circ + \mu_{\text{Cu}^{2+}}^\circ - \mu_{\text{Cu}}^\circ \} }_{=V_{oc}^\circ} + RT \ln \left(\frac{a_{\text{Cu}^{2+}} a_{\text{Zn}}}{a_{\text{Zn}^{2+}} a_{\text{Cu}}} \right) \\ V_{oc} &= V_{oc}^\circ + RT \ln \left(\frac{a_{\text{Cu}^{2+}}}{a_{\text{Zn}^{2+}}} \right) , \end{aligned}$$

as the activity of solid phases are equal to 1 by definition.¹ The activity of the dissolved ions are related (but not equal or proportional, as the behavior of a highly concentrated solution with charged ions deviates significantly from the ideal behavior) to the concentration of the ions at the interface close to the electrode, the regions where the reactions occur,

$$a_i = \gamma_i \frac{C_i}{C_i^\circ} .$$

Debye-Huckel theory provides a model for the activity coefficient γ_i , as a function of the ionic strength of the solution.

Ion	$\mu^\circ [\text{J/mol}]$	ϕ
$\text{Zn}^{2+}(aq)$	$-147.06 \cdot 10^3$	$\phi_a = -\frac{\mu_{\text{Zn}^{2+}}^\circ}{2F} = -0.762 \text{ V}$
$\text{Cu}^{2+}(aq)$	$65.49 \cdot 10^3$	$\phi_c = \frac{\mu_{\text{Cu}^{2+}}^\circ}{2F} = 0.340 \text{ V}$

The standard open-circuit cell potential of a Daniell cell is $V_{oc}^\circ = \phi_c^\circ - \phi_a^\circ = 1.102 \text{ V}$

Operation under loads

todo

Todo list

- write a full dynamical model
- write a simplified one, after removing any unnecessary dynamical variable

The goal is to write the constitutive laws of an electro-chemical component with approximately constant voltage and a state of charge internal variable.

¹ By definition the standard state of a solid is chosen to be the pure solid at the system temperature and a standard pressure. From this and Nernst equation, it immediately follows that $a = 1$ for the solid in the reference condition. From a more intuitive point of view, concentration of a solid doesn't change: adding a piece of a solid medium doesn't make the atoms in the crystal more packed, and thus the chemical environment around an atom of the solid lattice doesn't change, and thus its chemical potential neither.

Elements of the system

- **Electrode-electrolyte interfaces.**
 - activation overpotential
 - double-layer capacitance
- **Diffusion layer**
- **Ions migration**

Charge transfer kinetics

At the anode, for the law of mass action

$$\begin{aligned} v_a &= v_{a,f} - v_{a,r} = \\ &= A_{a,f} \exp\left(-\frac{E_{a,f}}{RT}\right) a_{\text{Zn}} - A_{a,r} \exp\left(-\frac{E_{a,r}}{RT}\right) a_{\text{Zn}^{2+}} = \end{aligned}$$

At the equilibrium, $v_a = 0$.

$$A_{a,f} \exp\left(-\frac{E_{a,f}^0}{RT}\right) a_{\text{Zn}}^0 = A_{a,r} \exp\left(-\frac{E_{a,r}^0}{RT}\right) a_{\text{Zn}^{2+}}^0 =: v_a^0.$$

In order to have a net velocity of reaction - out of the equilibrium - a net potential difference across the interface is required so that the energy levels of the reagents and the products shift, and so do the activation energies.

Activation overpotential is defined as the difference between the potential under load condition and the potential at the equilibrium

$$\eta_{act,a} = \phi_a - \phi_a^0.$$

The change in energy of a mole of ions due to the activation overpotential becomes $\Delta E_a = zF\eta_{act,a}$ and the activation energies for forward and reverse reactions become

$$\begin{aligned} E_{a,f} &= E_{a,f}^0 - \alpha_{a,f} \Delta E_a = E_{a,f}^0 - \alpha_{a,f} zF\eta_{act,a} \\ E_{a,r} &= E_{a,r}^0 + \alpha_{a,r} \Delta E_a = E_{a,r}^0 + \alpha_{a,r} zF\eta_{act,a} \end{aligned}$$

with $\alpha_{a,f} + \alpha_{a,r} = 1$. The velocity of the reaction becomes

$$\begin{aligned} v_a &= A_f \exp\left(-\frac{E_{a,f}}{RT}\right) - A_r \exp\left(-\frac{E_{a,r}}{RT}\right) = \\ &= A_f \exp\left(-\frac{E_{a,f}^0}{RT}\right) \exp\left(\frac{\alpha_{a,f} zF\eta_{act,a}}{RT}\right) - A_r \exp\left(-\frac{E_{a,r}^0}{RT}\right) \exp\left(-\frac{(1 - \alpha_{a,f}) zF\eta_{act,a}}{RT}\right) a_{\text{Zn}^{2+}}^0 \frac{a_{\text{Zn}^{2+}}}{a_{\text{Zn}^{2+}}^0}, \end{aligned}$$

i.e. the velocity of the reaction at the anode reads

$$v_a = v_a^0 \exp\left(\frac{\alpha_{a,f} zF\eta_{act,a}}{RT}\right) \left[1 - \frac{zF\eta_{act,a}}{RT} \frac{a_{\text{Zn}^{2+}}}{a_{\text{Zn}^{2+}}^0}\right].$$

In order to get a charge current, the velocity of reaction is multiplied by zF , i.e. zqN_A ,

$$i_a = v_a zqN_A,$$

as

$$\frac{\frac{[\text{charge}]}{[\text{time}]}}{i_a} = \frac{\frac{[\text{n.moles}]}{[\text{time}]}}{v_a} \underbrace{\frac{[\text{n.charges}]}{[\text{n.parts}]}}_z \underbrace{\frac{[\text{charge}]}{[\text{n.charges}]}}_q \underbrace{\frac{[\text{n.parts}]}{[\text{n.moles}]}}_{N_A}.$$

Extreme regimes

- Linear regime $\eta \ll \frac{RT}{zF}$, extremely rare
 - Tafel regime $\eta \gg \frac{RT}{zF}$, negative exponential of the reverse reaction collapses
-

12.2 Li-ion battery**12.2.1 Description****12.2.2 Regimes**

Open-circuit equilibrium

Operation under loads

Part IV

Random Notes

PROOF INDEX

definition-0

definition-0 (*ch/principles-gibbs-phase-rule*), 8

definition-1

definition-1 (*ch/heat-transfer-radiation*), 43

definition-2

definition-2 (*ch/principles-gibbs-phase-rule*), 9

definition-3

definition-3 (*ch/principles-gibbs-phase-rule*), 9

example-1

example-1 (*ch/principles-gibbs-phase-rule*), 8

example-10

example-10 (*ch/principles-gibbs-phase-rule*), 11

example-11

example-11 (*ch/principles-gibbs-phase-rule*), 11

example-12

example-12 (*ch/principles-gibbs-phase-rule*), 13

example-13

example-13 (*ch/principles-gibbs-phase-rule*), 13

example-14

example-14 (*ch/principles-gibbs-phase-rule*), 14

example-15

example-15 (*ch/principles-gibbs-phase-rule*), 14

example-16

example-16 (*ch/principles-gibbs-phase-rule*), 14

example-17

example-17 (*ch/principles-gibbs-phase-rule*), 14

example-18

example-18 (*ch/principles-gibbs-phase-rule*), 14

example-2

example-2 (*ch/heat-transfer-radiation*), 46

example-5

example-5 (*ch/principles-gibbs-phase-rule*), 10

example-6

example-6 (*ch/principles-gibbs-phase-rule*), 10

example-7

example-7 (*ch/principles-gibbs-phase-rule*), 10

example-8

example-8 (*ch/principles-gibbs-phase-rule*), 11

example-9

example-9 (*ch/principles-gibbs-phase-rule*), 11

proposition-4

proposition-4 (*ch/principles-gibbs-phase-rule*), 9

thermodynamics:principles:open:ex:comb-chamber

thermodynamics:principles:open:ex:comb-chamber (*ch/principles-open*), 17

thermodynamics:principles:open:ex:compressor

thermodynamics:principles:open:ex:compressor (*ch/principles-open*), 17

thermodynamics:principles:open:ex:turbine

thermodynamics:principles:open:ex:turbine (*ch/principles-open*), 17

thm-euler

thm-euler (*ch/potentials*), 19